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The study of water wettability on solid surfaces by molecular dynamics simulation☆

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ABSTRACT

The wetting of nanodroplets on solid surfaces was studied by molecular dynamics simulations. Through investigation of the contact angle of nanodroplets with different sizes, a relatively stable and accurate contact angle was obtained for a droplet radius of larger than 43 Å (1 Å = 10^{-10} m). The contact angles for droplets at different levels of solid-liquid forces were quantitatively evaluated. The results show that there is an excellent linear correlation between them in the angle range of 28° to 167°. Under a strong solid-liquid force, thin liquid film expansion in the three-phase line was found in nanometer-level simulations. The thin film spreading range increased with increasing solid-liquid force.

1. Introduction

Water is the most common substance on Earth, and an essential component indispensable for life. In the past 40 years, many studies have been conducted on water, including its physical and chemical properties [1], disclosed the mechanisms underlying the special behavior of water droplets on solid surfaces [2], and set off a wave of nanosurface applications [3]. Some superhydrophobic [4–5] and superhydrophilic [6–7] surfaces have been developed and used in production and life. Since it is difficult to observe the movement of nano-level water molecules, molecular dynamics (MD) simulations have been used to study the properties of water molecules in recent years. Many potential energy models of water molecules have been developed for molecular dynamics, such as TIP3P [8], TIP4P [9], TIP5P [10], and SPC/E [11]. In the past 20 years, numerous simulations of the behavior of nano water droplets have been performed, including the behavior of droplets on the surface of nanostructures [12–16] and new materials such as graphene [17–20],and special expansion of the thin liquid film at three-phase line [21–24].

Due to the different forces between water-water and water-solid molecules, when water droplets are placed onto different solid surfaces, they will show different contact angles [25–29]. In the Lennard-Jones potential functions, the interaction potential ε between the solid and liquid plays the most important role in the contact angle alteration [19,29-30]. The increase in ε enhances the water-solid force, leading to the adsorption of droplets onto the solid wall to form a smaller contact angle. For instance, T. Werder et al. [19] increased ε from 0.1881 kJ/mol to 0.6270 kJ/mol when water droplets were placed onto the surface with a graphene structure sheet and observed that the contact angle decreased monotonically with increasing ε, and there was a linear relationship between them. Monojit Chakraborty et al. [29] reached a similar conclusion by placing water droplets containing 12000 molecules onto a solid surface and found that the contact angle decreased from 145° to 10° when ε increased from 0.10 kJ/mol to 4.02 kJ/mol. Moreover, Seung do Hong [30] also performed similar work and obtained similar conclusions as above.

On the other hand, the sizes of the droplets will also affect the contact angle. Compared with a water droplet composed of 16,000 molecules with a contact angle of 67°, the size of a droplet composed of 1,024,000 molecules will increase by approximately 10% to 74° [31]. In the case of the nanolevel, the effect of water droplet size on gold [32], kaolinite [33], or graphite substrates [34] is the same, and, for all these cases, the contact angle increases with increasing droplet size.

Moreover, a precursor film [26,35–36] was found at the position of

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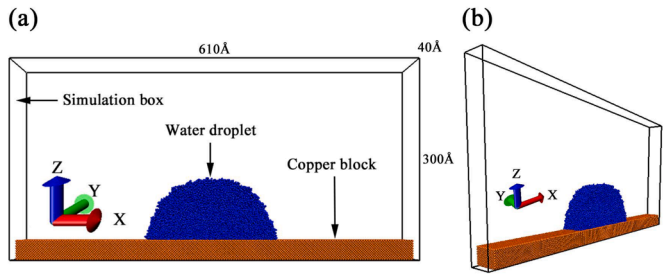


Fig. 1. Schematic diagram of simulation system coordinates: (a) Front view (b) Oblique view.

the three-phase line when droplets were spread on a solid surface. The precursor film was also called a thin liquid film [37]. In addition to typical water wetting, this kind of thin liquid film also appears in nonvolatile "dry spreading" [38] and refrigerants [39]. This phenomenon mainly occurs on a hydrophilic surface. The contact line region have been researched in recent years through the use of advanced nano experiments and molecular dynamics simulations [21–24]. In the wetting dynamics, the expansion of the thin liquid film in the contact line can be described by the power function [24]. In this paper, the density and shape of thin liquid film in the contact line region are similar to those of previous studies.

However, the number of water molecules used in molecular dynamics simulations in existing studies is generally approximately several hundred to several thousand. The molecules on the droplet contour will make complex oscillating motion under the combined action of interaction force and molecular thermal movement . Due to the small number of molecules used, the irregular thermal movement of molecules on the boundary causes instability of the boundary, which will increase the fitting deviations of droplet's radius and contact angle. Therefore, we study the effect of instability on radius and contact angle, and obtains the number needed to form a stable droplet. Simulation show that when the radius of nanodroplets is above 43 Å, a relatively stable circular boundary can be formed. To further reduce the resulting deviations, droplets consisting of tens of thousands of water molecules are placed onto a copper surface, and the precise contact angle of droplets is calculated utilizing density segmentation and boundary fitting. Furthermore, the influence of droplet size on measurement of the contact angle and the thin liquid film at different hydrophilicity surfaces is discussed in this article.

2. Simulations Details

In this paper, we used the molecular dynamics software LAMMPS [40] to predict the behavior of liquid nanodroplets. LAMMPS's code supports parallel operations [41], which can speed up the calculation process. The data visualization software uses visual molecular dynamics (VMD) [42], which can be used to observe changes in water droplet shape and collect coordinate information for molecules. Considering that for water droplets, gravity is much less important than the surface stress at the nanoscale, the effect of gravity was ignored. The interaction potential between two molecules uses the classical Lennard-Jones potential, and the electric fields forces between atoms use the long-range coulombic interaction potential. The Lennard-Jones potential is written as follows

$$E_{ij}(r) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \quad (1)$$

The long-range coulombic interactions potential is given by

$$E_e(r) = C\frac{q_i q_j}{r_{ij}} \quad (2)$$

where ε describes the interaction potential between two molecules, and σ is the distance between the equilibrium positions of the two molecules. r is the distance between two molecules, the subscripts i and j represent two different atoms, and C is the constant for the coulombic interactions. Lennard-Jones potential is used between different electronal neutral atoms to describe the interaction potential between them. Due to the dipole moment of water molecules, the long-range coulombic interaction potential is added to describe the potential between water molecules, especially for SPC/E (extended simple point charge) [43] water molecules.

2.1. Simulation box

The box size is set to 610 × 40 × 300 Å (XYZ, 1 Å = 10^{-10} m), as shown in Fig. 1. Then, the PPP boundary is used in the simulation box, where p indicates that the box boundary is periodic. In this case, the forces between atoms can cross the boundary, and atoms can exit one boundary of the box and re-enter the box from another boundary. A great advantage of using the PPP boundary is that a small number of molecules can be used to simulate larger droplets [15], thus greatly saving computational resources, and there are definite three-phase lines and contact angles in the model. To ensure enough data for density division and reduce the number of molecules, the thickness in the Y direction is set to 40 Å. Another advantage of using the PPP boundary is that there is a wireless solid-liquid contact line in the Y direction, which can avoid the influence of line tension [44] on the contact angle. The line tension of the three-phase balancing system is defined as the free energy at the three-phase contact line and the length tends to be minimized. According to the research [45], it is related to the curvature radius of the droplet, which has a great influence on the contact angle of nanodroplets. Due to the use of PPP boundary, the three-phase line in the model is a straight line with wireless length and will not shrink, which avoid the influence of line tension.

2.2. Water

SPC/E water molecules were used in the simulation.The cutoff distance L_{cut} for the calculation of the L-J potential and the long-range coulombic interaction potential between molecules was set to 9 Å. A total of 18,000 water molecules were set up in the simulation. The ensemble of water was set to NVT, where the temperature was set to 298.15 K through Nose-Hoover [46–47], a thermal bath can stabilize the temperature at the set value.

2.3. Copper

A copper block was set at the bottom of the box, whose size was 590 × 40 × 30 Å. The arrangement of copper atoms was FCC face-centered cubic, with a lattice constant of 3.61 Å. To simplify the calculation, no force was set between copper atoms [19], which means that the simulation process will not perform calculations on copper. The cutoff distance between copper and oxygen atoms was 10 Å, which is smaller than the thickness of copper in the Z direction of 30 Å, so the influence of copper thickness can be ignored. The L-J potential was used to describe the force between copper and oxygen atoms. ε_{O-Cu} and σ_{O-Cu} can be obtained through the Lorentz-Bertholet mixing rule [48]:

$$\varepsilon_{O-Cu} = \sqrt{\varepsilon_{O-O} * \varepsilon_{Cu}} \quad (3)$$

$$\sigma_{O-Cu} = \frac{\sigma_{O-O} + \sigma_{Cu}}{2} \quad (4)$$

where the subscripts O and Cu represent oxygen and copper atoms, respectively. We set the simulation duration to 2 ns to ensure that the system reached a stable state, where 1ps for one step, and a total of 2 million steps. Then, use the LAMMPS to output the coordinates file of all molecules every 0.001 ns, and the software VMD was used for visuali-

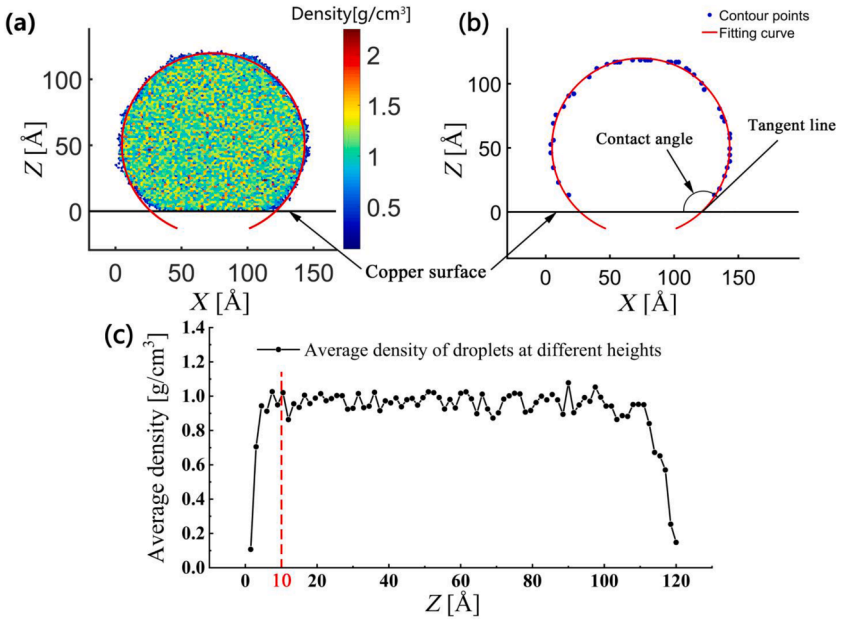


Fig. 2. Contact angle measurement: (a) Water droplet density map (b) Circular fitting curve (c) Average density of droplets at different heights.

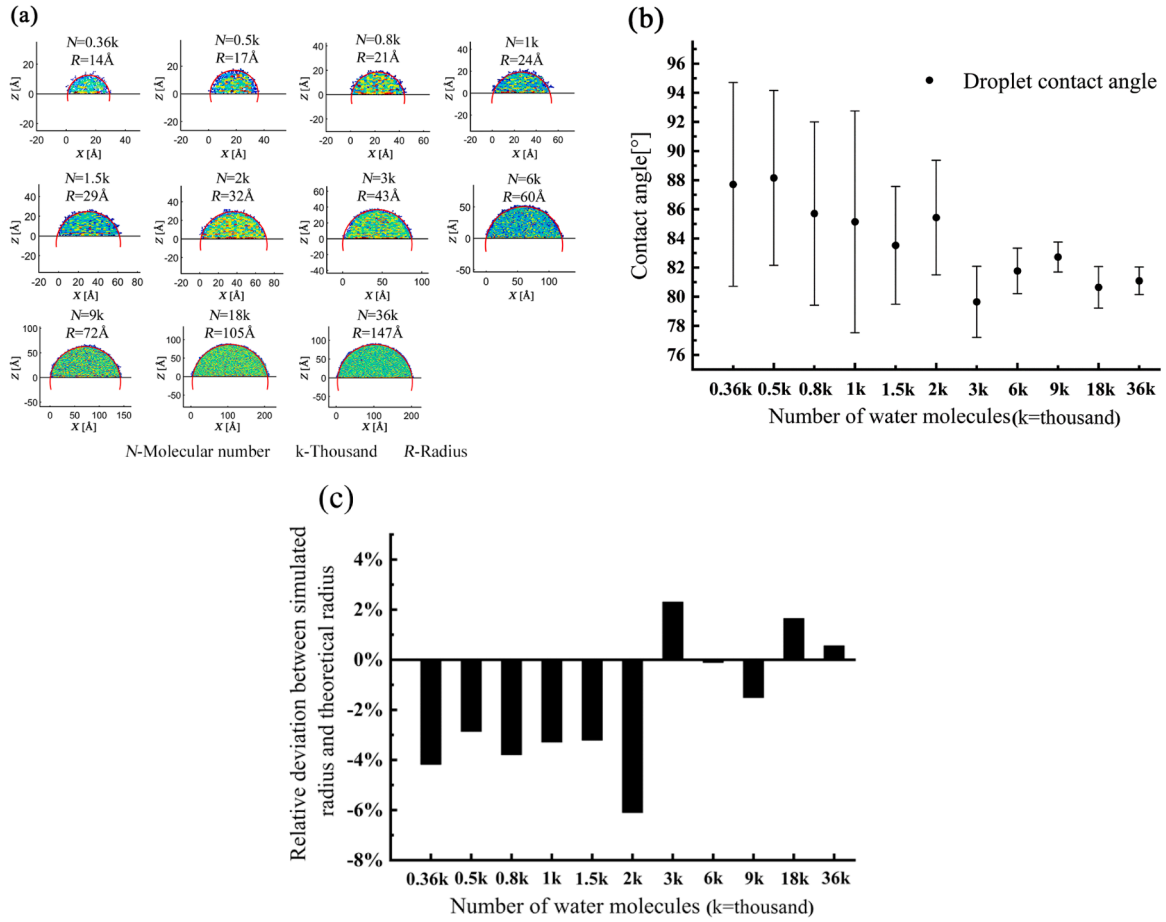


Fig. 3. Droplets of different sizes: (a) Circle fitting of profiles for differently sized droplets, (b) Contact angle measured with different droplet sizes, and (c) Relative deviation between simulated radius and theoretical radius.

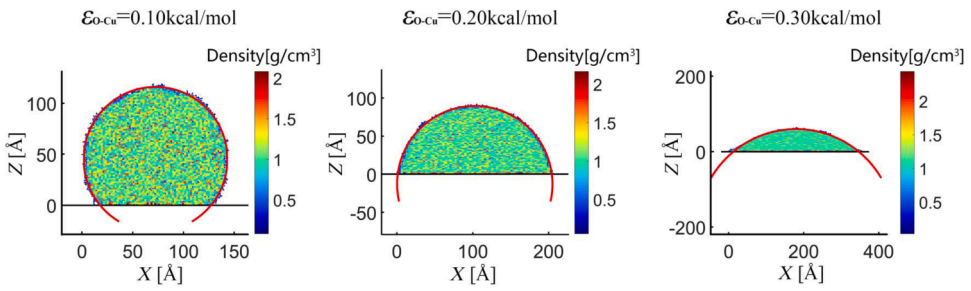


Fig. 4. Contact angle for varying ε_{O-Cu}.

zation and coordinate collection.

2.4. Contact angle measurement

Two main methods are used to calculate the contact angle in simulations [49]. One is to use the interfacial tension (γ) between the solid phase (S), liquid phase (L), and gas phase (V) to calculate the contact angle (θ) based on Young's equation [50–51]:

$$\cos\theta = (\gamma_{SV} - \gamma_{SL})/\gamma_{LV} \quad (5)$$

The other way involves dividing the cross-section of the droplets into many regions and calculating the density of each region [17-18, 30]. Then, one can identify the region with a certain density (generally half the density of water) as the boundary of the droplets. Next, the boundary is fitted by an arc, so the angle between the arc and the solid surface is the contact angle.

The second method is used in this paper. First, a thickness of 40 Å in the Y direction was regarded as a plane, and all molecules were projected onto the X-Z plane. Then, the density division method was used to divide the water droplets into many different regions, and an area with a density of 0.5 g/cm^3 was defined as the outer profile of the droplets, as shown in Fig. 2(a) (ε_{O-Cu} = 0.07kcal/mol). In the absence of gravity, the droplets will form a sphere due to surface tension, which means that the profile of the droplets' cross-sections will be a circular arc. Next, the points on the profile were removed to fit a circular arc, which is shown in Fig. 2(b). As shown in Fig. 2(c), due to the Lennard-Jones potential between the solid and liquid, there will be repulsive or attractive force between solid-liquid atoms at close range, resulting in a sudden change in the water density at a distance of 10 Å near the wall. So it is worth mentioning that the region near the copper surface 10 Å was excluded to avoid the influence of density change [19]. But the low density on the droplet contour makes the value on the right side of the curve in Fig. 2(c) smaller, which has no effect on curve fitting. Then, the slope of the arc at the three-phase line was calculated to obtain the contact angle.

In the process of simulation, the droplets will have some irregular small motion in the equilibrium state, and the arc fitting greatly improves the stability of the contact angle prediction. For an experiment, 10 frames were taken at different times to fit the contact angle of a state, and we took the average value as its contact angle.

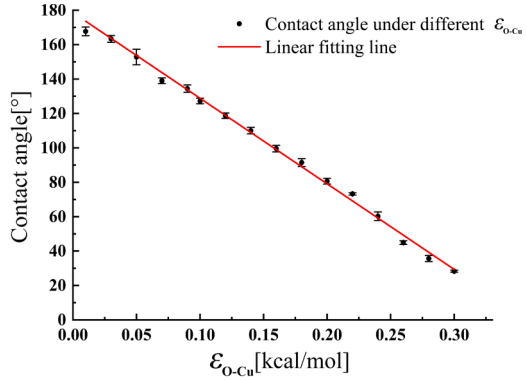


Fig. 5. Contact angle under varying ε_{O-Cu}.

3. Results and discussion

3.1. Effect of droplet size on simulation

In nanodroplet contact angle MD simulations, the number of molecules is an important parameter that needs to be determined. A too large number of molecules will slow down the calculation speed, but too few molecules will lead to an unstable droplet shape and can even make it difficult to form a stable profile. To investigate the influence of droplet size on the contact angle in molecular simulation, 11 water droplets with different numbers ranging from 360 to 36,000 were set in simulations. The thickness in the Y direction remained 40 Å, ε_{O-Cu} was set to 0.2 kcal/mol, and the temperature was 298.15 K. It is worth noting that for the universality of the simulation, copper is regarded as a solid with a special crystal structure here. So ε_{O-Cu} = 0.2kcal/mol is only selected and used to evaluate the simulation results effectively. After the simulation system reached equilibrium, each simulation randomly collected 10 stable frames to calculate the contact angle.

Fig. 3(a) shows the droplet profile formed by different numbers of molecules, and we found that even if the small droplets reached a stable state after a sufficiently long time, the shape still showed large random fluctuations. The fluctuation sizes of different nano droplets at the boundary is similar under thermal motion. Therefore, for smaller size droplets, the ratio of fluctuation size to droplet size will be larger, and the random deformation of droplets will be more obvious. First of all, it's obvious that shape instability is reflected in the calculation of the contact angle. As shown in Fig. 3(b), it can be seen that the contact angles of different sizes are almost the same in the studied range, but the numerical deviation calculated with a small size is much larger. For example, when the number of molecules is less than 2000, which means that the radius of the droplet is less than 32 Å, the standard deviation of the measured contact angle is more than 4°. However, when the number of molecules is more than 3000, the radius of the droplet is more than 43 Å, the standard deviation of the contact angle will be less than 2.5°, and the relative deviation can be controlled within 3%.

Such deviation also reflects on the measured radius. In order to evaluate the credibility of the radius obtained by simulation, the contact angle of the droplet formed by 36,000 water molecules is defined as the standard value when ε_{O-Cu} is 0.2 kcal/mol. Next the theoretical radius of droplet with different molecular number is calculated by using the density of water in normal temperature, and then compared with the simulation results. The relative deviations between the simulated radius and the theoretical radius are shown in Fig. 3(c), which agree with each other and show the reliability of the simulation. It shows that as the number of molecules increases to more than 3,000, the relative deviation of the radius will be significantly reduced.

Overall, to obtain a more stable and accurate contact angle, the molecular number of droplets is suggested to be set to more than 3,000. The droplet radius of this size is approximately 43 Å, which is

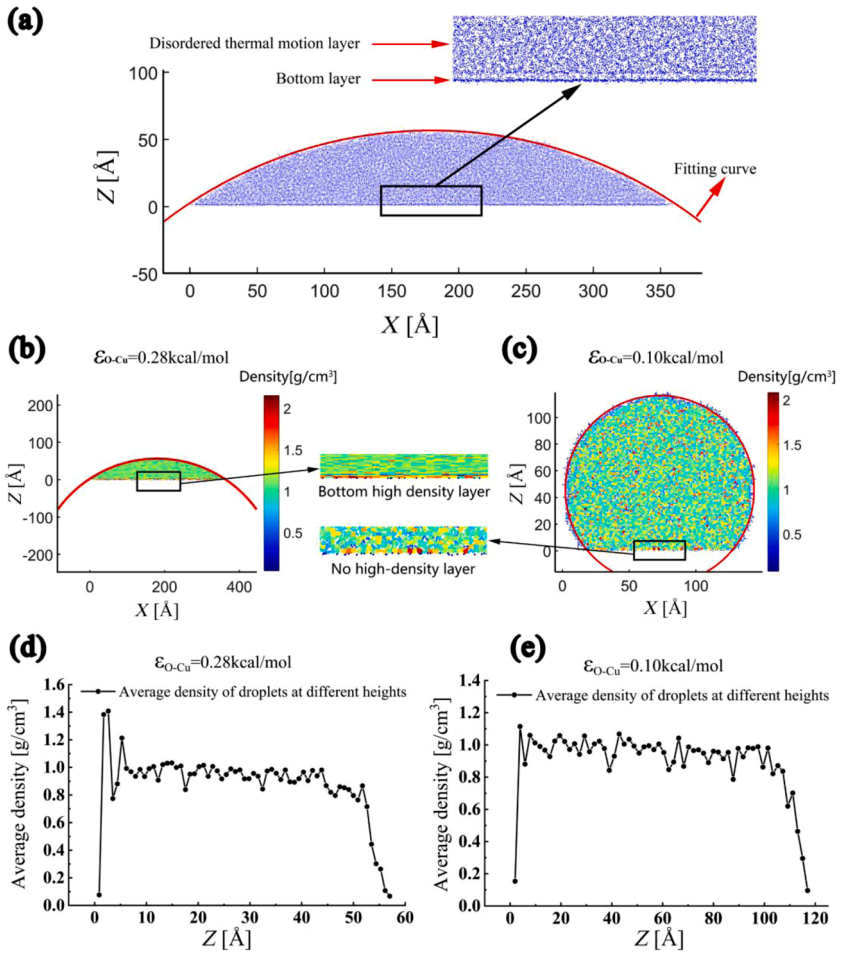


Fig. 6. The high-density layer at the bottom formed by the force of solid and liquid: (a) A layer is formed at the bottom under strong ε_{S-L}, (b) The high-density layer at the bottom is formed under the strong ε_{S-L}, (c) No obvious high-density layer formed on the bottom under weak ε_{S-L}, (d) Average density of droplets at different heights under the strong ε_{S-L}, and (e) Average density of droplets at different heights under the weak ε_{S-L}.

approximately 4 times the cutoff distance of the L-J potential between water and copper.

3.2. Contact angle under different ε_{S-L} (S-Solid, L-Liquid)

The wetting system composed of copper and water used in the simulation is a special solid-liquid system, which possess Lennard-Jones potential between solid and liquid. In this case, copper can be consider as an arbitrary solid, and different ε_{O-Cu}(ε_{S-L}) can be set in simulation to investigate its influence on the contact angle.

In this part, 20 experiments for varying ε_{O-Cu} were set from 0.01 kcal/mol to 0.40 kcal/mol. The interaction forces in the simulations included the force between different water molecules and the force between water molecules and copper atoms. As shown in Fig. 4 , the contact angle decreases as ε_{O-Cu} increases. When the value of ε_{O-Cu} is small, the internal cohesion of the water molecules plays a major role, resulting in a larger contact angle. When the value of ε_{O-Cu} is large, the adsorption of copper on the water molecules near the bottom surface is great, the water molecules are adsorbed onto the copper surface, and the force between the solids and liquid plays a major role in the shape of the droplet and forms a smaller contact angle. For example, with increasing ε_{O-Cu} from 0.01 kcal/mol to 0.30 kcal/mol, the contact angle of the water droplets decreases from 167° to 28°. Our simulation changing trend are consistent with existing research [29].

A total of 18,000 water molecules were used to measure the contact angle of the droplets. After a relaxation time of 2 ns in the simulation, the droplet boundary shape was stable, and the arc fit can be used to describe the shape of the droplet well. The droplet contact angle varies with ε_{O-Cu}, which is shown in Fig. 5. When ε_{O-Cu} exceeds 0.30 kcal/mol, due to the strong adsorption of the copper surface, a large expansion of water droplets will occur at both ends of the solid-liquid contact surface, which will be discussed later. Therefore, within the range of good arc fitting, the contact angle in the case of ε_{O-Cu}=0.01~0.30 kcal/mol is selected for fitting a line. It is impressive that ε_{O-Cu} and the contact angle conform to a linear relationship, i.e.,

$$\theta = 178.57^{\circ} - -497.41^{\circ} \times \varepsilon_{O-Cu}. \quad (6)$$

where θ is the droplet contact angle. The correlation coefficient in linear fitting is 0.995, which shows excellent predictive power. We can also observe similar relationships in some researchs [18,29,52], which show the contact angle decreases as the interaction potential increases.

3.3. Thin liquid film extended distance under strong ε_{S-L} (S-Solid, L-Liquid)

When liquid molecules are close to the solid surface, they are mainly subjected to the force of the solid. As shown in Fig. 6(a), when ε_{S-L} is large, the solid has strong adsorption to the liquid, and the liquid molecules near the solid appear to have a separate layered distribution. The molecular thermal motion of this layer is also limited by the solid-liquid force, so liquid molecules have difficulty leaving the solid surface. However, the upper liquid molecules still undergo irregular thermal movement and are not significantly affected by solids.

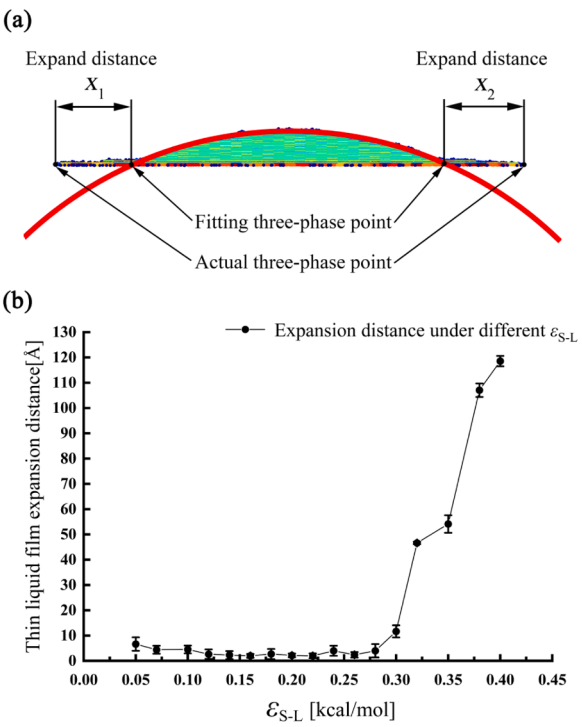


Fig. 7. Extended distance of thin liquid film under strong ε_{S-L}: (a) Extended distance between fitted and actual three-phase points, (b) Extended distance under varying ε_{S-L}.

Fig. 6(b) shows the density contour of the droplet when ε_{S-L} is 0.28 kcal/mol and Fig. 6(d) shows the average density of droplets at different heights in this case. As shown in Fig. 6(d), with the increase of *Z*, the liquid density will increase from a small value to a value beyond the normal density, then back to normal until it rises to the low density contour of the droplet. Under the strong solid-liquid force, the liquid density of the absorption layer close the solid surface is approximately 1.4 g/cm^3, while the density of upper water is 1 g/cm^3. High density delamination at the bottom occurs on the whole solid liquid contact surface. However, this phenomenon does not occur when ε_{S-L} is weak, as shown in Fig. 6(c) and Fig. 6(e).

It is also found that when ε_{S-L} is larger than 0.32 kcal/mol, the droplets will have an expanded thin liquid film [37] formed on both sides of the X direction, which have similar size and shape. As shown in Fig. 7(a), there is a large gap between the three-phase point predicted by arc fitting and the actual three-phase point formed by the extended liquid film on both sides. To characterize the size of the expanded thin liquid film, the extended distance between the fitted three-phase point and the actual three-phase point was defined as the extended distance of the thin liquid film (*X*, the average of X_1 and X_2 in Fig. 7(a)).

Simulations of different ε_{S-L} values ranging from 0.05 kcal/mol to 0.40 kcal/mol were performed to investigate the extended distance of the thin liquid film, as plotted in Fig. 7(b). When ε_{S-L} is less than 0.30 kcal/mol, the fitting three-phase point and actual three-phase point almost overlap, and the thin liquid film extended distance on both sides is less than 10 Å, so it can be considered that no thin liquid film exists. However, when ε_{S-L} is above 0.30 kcal/mol, the gap between the fitting and actual three-phase points increases rapidly, leading to a significant increase in the extended distance. When ε_{S-L} is 0.32 kcal/mol, the extended distance is 46 Å, and when ε_{S-L} is 0.40 kcal/mol, the extended distance will reach 107 Å.

The formation of the extended thin liquid film is mainly due to the strong solid-liquid interaction, which leads to attraction of liquid molecules near the solid to the solid wall. When ε_{S-L} is 0.30 kcal/mol, the corresponding contact angle is 28°. Therefore, it is believed that when the solid-liquid contact angle is less than this angle, there will be an obvious existence of the extended thin liquid film; the smaller the contact angle is, the larger the extended distance of the thin liquid film will be.

4. Conclusions

In this paper, we used molecular dynamics to simulate the wetting of nano water droplets on a solid surface. To explore the influence of droplets with different sizes on contact angle formation, we set up simulation experiments for several different droplet sizes. The results indicate that the contact angles for droplets with different sizes ranging from 360 to 36,000 molecules are approximate. However, the unstable fluctuation of the profile in small droplets affects the measurement of the contact angle, which is distinct when the radius is less than 32 Å. In contrast, more accurate contact angle values were obtained when more than 3,000 molecules were used to form droplets with a corresponding radius of 43 Å. The stability of the contact angle measurement was also found to be much better.

Different potentials for solid and liquid molecules (ε_{S-L}) will form different contact angles on the solid surface. The simulation results show that the contact angle in the range of 28° to 167° is linearly related to ε_{S-L}. The contact angle decreased with increasing ε_{S-L}. Moreover, when the droplets on a solid surface had a strong ε_{S-L}, a thin liquid film extension was found to be formed at the site of the three-phase line. An obvious thin-film extension was observed when the contact angle was less than 28°.

CRediT authorship contribution statement

Yinhao Yu: Conceptualization, Methodology, Investigation, Data curation, Writing – review & editing, Writing – original draft. **Xiongwen Xu:** Supervision, Project administration. **Jinping Liu:** Supervision. **Yuehui Liu:** Investigation, Data curation. **Wenhao Cai:** Data curation. **Jianxun Chen:** Investigation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.susc.2021.121916.

References

[1] B. Guillot, J. Mol. Liq. 101 (2002) 219.
[2] W. Barthlott, C. Neinhuis, Purity of the sacred lotus, or escape from contamination in biological surfaces, Planta 202 (1997) 1–8.
[3] H.F. Bohn, W. Federle, Insect aquaplaning: Nepenthes pitcher plants capture prey with the peristome, a fully wettable water-lubricated anisotropic surface, PNAS 101 (2004) 14138–14143.
[4] A. Marmur, S. Kojevnikova, Super-hydrophobic surfaces: Methodological considerations for physical design, Journal of Colloid and Interface Science 568 (2020) 148–154.

[5] S. Khorsanda, K. Raeissi, F. Ashrafizadeh, M.A. Arenas, A. Conde, Corrosion behaviour of super-hydrophobic electrodeposited nickel–cobalt alloy film, Appl. Surf. Sci. 364 (2016) 349–357.
[6] J. Zhao, et al., A rapid one-step nanosecond laser process for fabrication of super-hydrophilic aluminum surface, Optics & Laser Technology 117 (2019) 134–141.
[7] J. Kang, et al., Subcooled water quenching on a super-hydrophilic surface under atmospheric pressure, International Journal of Heat and Mass Transfer 117 (2018) 538–547.
[8] W.L. Jorgensen, et al., Comparison of simple potential functions for simulating liquid water, The Journal of Chemical Physics 79 (2) (1983) 926–935.
[9] W.L. Jorgensen, J.D. Madura, Temperature and size dependence for Monte Carlo simulations of TIP4P water, Molecular physics 56 (6) (2006) 1381–1392.
[10] M.W. Mahoney, W.L. Jorgensen, A five-site model for liquid water and the reproduction of the density anomaly by rigid, nonpolarizable potential functions, The Journal of Chemical Physics 112 (20) (2000) 8910–8922.
[11] H.J.C. Berendsen, J.R. Grigera, T.P. Straatsma, J. Phys. Chem. 91 (1987) 6269.
[12] H. Li, K. Zhang, Dynamic behavior of water droplets impacting on the superhydrophobic surface: Both experimental study and molecular dynamics simulation study, Applied Surface Science 498 (2019), 143793.
[13] T. Koishi, et al., Measurement of Contact-Angle Hysteresis for Droplets on Nanopillared Surface and in the Cassie and Wenzel States: A Molecular Dynamics Simulation Study, ACS Nano 5 (9) (2011) 6834–6842.
[14] M.S. Ambrosia, J. Jang, M.Y. Ha, Static and dynamic hydrophobicity on a nano-sized groove/ridge surface, Computers & Fluids 114 (2015) 75–83.
[15] A. Thomas, N.V. Priezjev, Molecular dynamics simulation study of a polymer droplet transport over an array of spherical nanoparticles, Computational Materials Science 184 (2020), 109872.
[16] H. Li, et al., Dynamic Contact Angles and Mechanisms of Motion of Water Droplets Moving on Nanopillared Superhydrophobic Surfaces: A Molecular Dynamics Simulation Study, Langmuir 34 (34) (2018) 9917–9926.
[17] D. Sergi, G. Scocchi, A. Ortona, Molecular dynamics simulations of the contact angle between water droplets and graphite surfaces, Fluid Phase Equilibria 332 (2012) 173–177.
[18] R. Burt, et al., Molecular Dynamics Simulations of the Influence of Drop Size and Surface Potential on the Contact Angle of Ionic-Liquid Droplets, The Journal of Physical Chemistry C 120 (28) (2016) 15244–15250.
[19] T. Werder, et al., On the Water−Carbon Interaction for Use in Molecular Dynamics Simulations of Graphite and Carbon Nanotubes, The Journal of Physical Chemistry B 107 (6) (2003) 1345–1352.
[20] J. Rafiee, et al., Wetting transparency of graphene, Nature Materials 11 (3) (2012) 217–222.
[21] Y. Weng, et al., Spreading dynamics of a precursor film of nanodrops on total wetting surfaces, Physical chemistry chemical physics: PCCP 19 (4) (2017) 27786–27794.
[22] Q. Yuan, Y. Zhao, Precursor film in dynamic wetting, electrowetting, and electro-elasto-capillarity, Physical review letters 104 (24) (2010), p. 246101-246101.
[23] R.E. Isele-Holder, A.E. Ismail, Classification of precursors in nanoscale droplets, Physical review. E 93 (4) (2016), p. 043319-043319.
[24] H. Wang, From Contact Line Structures to Wetting Dynamics, Langmuir 35 (32) (2019) 10233–10245.
[25] E.B. Dussan, On the Spreading of Liquids on Solid Surfaces: Static and Dynamic Contact Lines, Annual review of fluid mechanics 11 (1) (1979) 371–400.
[26] P.G. de Gennes, Wetting: statics and dynamics, Rev. Mod. Phys. 57 (1985) 827. –Published 1 July.
[27] J.U. Brackbill, D.B. Kothe, C. Zemach, A continuum method for modeling surface tension, Journal of Computational Physics 100 (2) (1992) 335–354.
[28] D.Y. Kwok, A.W. Neumann, Contact angle measurement and contact angle interpretation, Advances in Colloid and Interface Science 81 (3) (1999) 167–249.
[29] M. Chakraborty, et al., Molecular Dynamics Study of Thermally Augmented Nanodroplet Motion on Chemical Energy Induced Wettability Gradient Surfaces, Langmuir 31 (41) (2015) 11260–11268.
[30] S.D. Hong, M.Y. Ha, S. Balachandar, Static and dynamic contact angles of water droplet on a solid surface using molecular dynamics simulation, Journal of Colloid and Interface Science 339 (1) (2009) 187–195.
[31] E. Santiso, C. Herdes, E. Müller, On the Calculation of Solid-Fluid Contact Angles from Molecular Dynamics, Entropy 15 (12) (2013) 3734–3745.
[32] C. Wu, et al., Effects of temperature, size of water droplets, and surface roughness on nanowetting properties investigated using molecular dynamics simulation, Computational Materials Science 53 (1) (2012) 25–30.
[33] R. Šolc, et al., Wettability of kaolinite (001) surfaces — Molecular dynamic study, Geoderma 169 (2011) 47–54.
[34] D. Sergi, G. Scocchi, A. Ortona, Molecular dynamics simulations of the contact angle between water droplets and graphite surfaces, Fluid Phase Equilibria 332 (2012) 173–177.
[35] W.B. Hardy III, The spreading of fluids on glass, The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science 38 (223) (2009) 49–55.
[36] D. Bonn, et al., Wetting and spreading, Reviews of Modern Physics 81 (2) (2009) 739–805.
[37] Cazabat, A.M., et al., Dynamics of Wetting: From Theory to Experiment. Interface Science. 5(2/3): p. 129-139.
[38] A.-M. Cazabat, F. Heslot, P. Carles, S.M. Troian, Hydrodynamic fingering instability of driven wetting films, J. Colloid Interface Sci. 39 (1992) 61.
[39] X. Lu, et al., Change in wetting characteristic of heated refrigerant in falling film evaporation, International Journal of Refrigeration 117 (2020) 198–208.
[40] R. Paula Leite, M. de Koning, Nonequilibrium free-energy calculations of fluids using LAMMPS, Computational Materials Science 159 (2019) 316–326. Available at, https://lammps.sandia.gov.
[41] S. Plimpton, Fast Parallel Algorithms for Short-Range Molecular Dynamics, Journal of Computational Physics 117 (1) (1995) 1–19.
[42] S. Cross, et al., Visualisation of cyclic and multi-branched molecules with VMD, Journal of Molecular Graphics and Modelling 28 (2) (2009) 131–139.
[43] H.J.C. Berendsen, J.R. Grigera, T.P. Straatsma, The missing term in effective pair potentials, The Journal of Physical Chemistry 91 (24) (1987) 6269–6271.
[44] B.M. Law, et al., Line tension and its influence on droplets and particles at surfaces, Progress in Surface Science 92 (1) (2017) 1–39.
[45] X. Ma, J. Lei, J. Xu, Line Tension of Nanodroplets on a Concave Surface, Langmuir 37 (15) (2021) 4432–4440.
[46] S. Nose, A Molecular dynamics method for simulations in the canonical ensemble, Mol. Phys. 52 (1984) 255–268.
[47] W.G. Hoover, Canonical dynamics: Equilibrium phase-space distributions, Phys. Rev. A 31 (1985) 1695–1697.
[48] D. Boda, D. Henderson, The effects of deviations from Lorentz–Berthelot rules on the properties of a simple mixture, Molecular Physics 106 (20) (2008) 2367–2370.
[49] H. Jiang, A.J. Patel, Recent advances in estimating contact angles using molecular simulations and enhanced sampling methods, Current Opinion in Chemical Engineering 23 (2019) 130–137.
[50] H. Jiang, F. Müller-Plathe, A.Z. Panagiotopoulos, Contact angles from Young's equation in molecular dynamics simulations, The Journal of Chemical Physics 147 (8) (2017), 084708.
[51] J. Fan, et al., Microscopic Origin of Capillary Force Balance at Contact Line, Phys Rev Lett 124 (12) (2020), 125502.
[52] M.R. Stukan, P. Ligneul, J.P. Crawshaw, E.S. Boek, Spontaneous Imbibition in Nanopores of Different Roughness and Wettability, Langmuir 26 (16) (2010) 13342–13352.